

Customer Behavior Analysis

Transforming Retail Insights: A Data Engineering and Visualization Journey via
Python, SQL, and Power BI

EXECUTIVE OVERVIEW

- ✓ **Cleaned Dataset:** Processed 5,000+ transaction records with zero data leakage.
- ✓ **Revenue Performance:** Achieved total market revenue of \$965.81K across 5 categories.
- ✓ **Loyalty Backbone:** 70%+ of the customer base identified as "Loyal" (>10 purchases).
- ✓ **Key Segment:** The 51+ age demographic is the primary revenue engine (\$387.6K).
- ✓ **Logistics ROI:** Express shipping drives the highest Average Order Value (\$345.07).

CORE BUSINESS ISSUES



Low Retention

Opaque customer purchase cycles making it difficult to predict and prevent churn.



Weak Segmentation

One-size-fits-all marketing resulting in missed opportunities in high-value niches.



Margin Erosion

Aggressive seasonal discounting without measuring actual impact on total revenue.

PYTHON DATA CLEANING

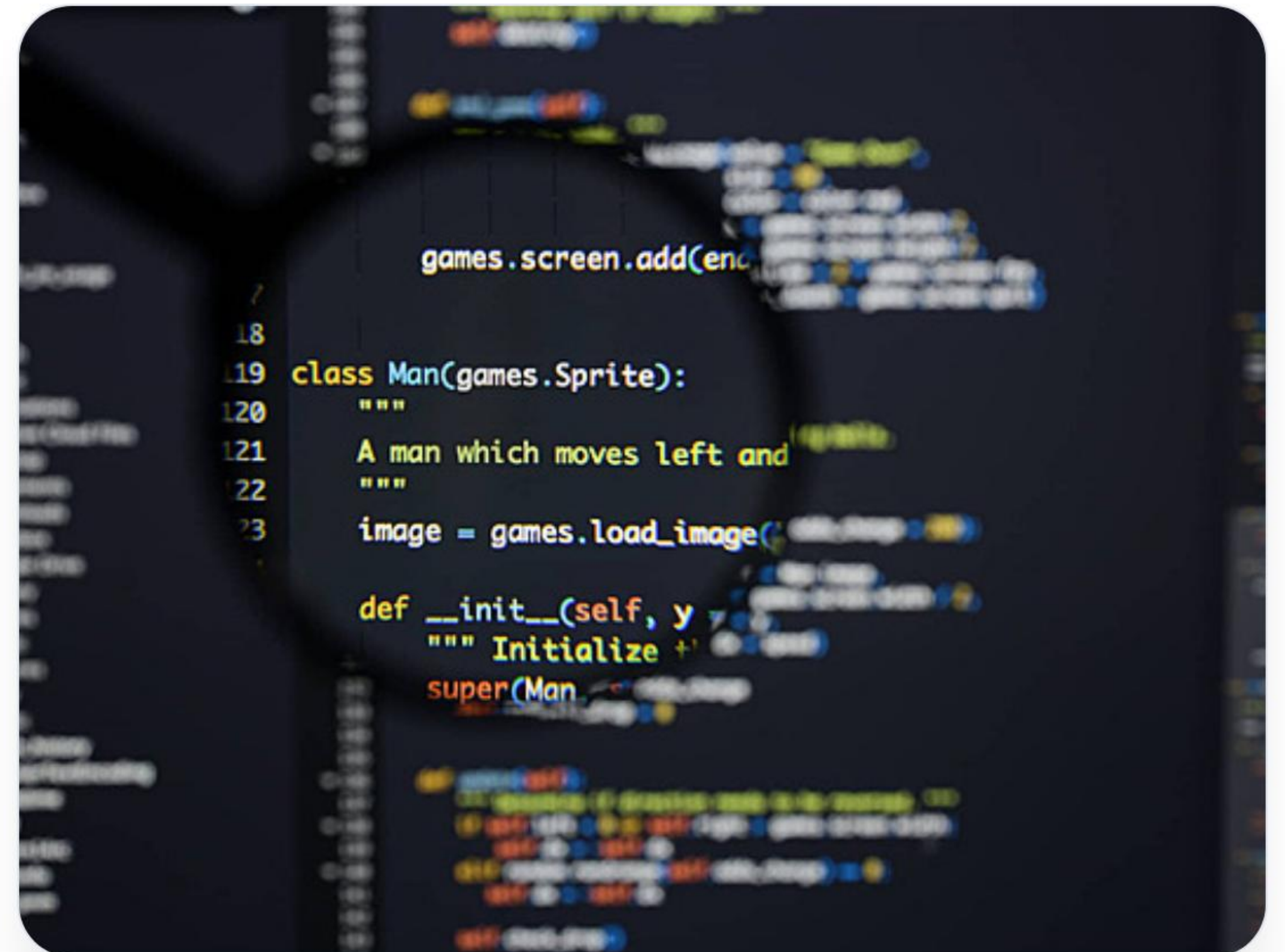
Building the Foundation

Cleaned and transformed the data using Pandas to ensure analytical readiness.

Categorization: Automated mapping of SKU items to departments.

Imputation: Applied median review ratings to handle null fields.

Database Sync: Engineered a SQLAlchemy export pipeline to MySQL.

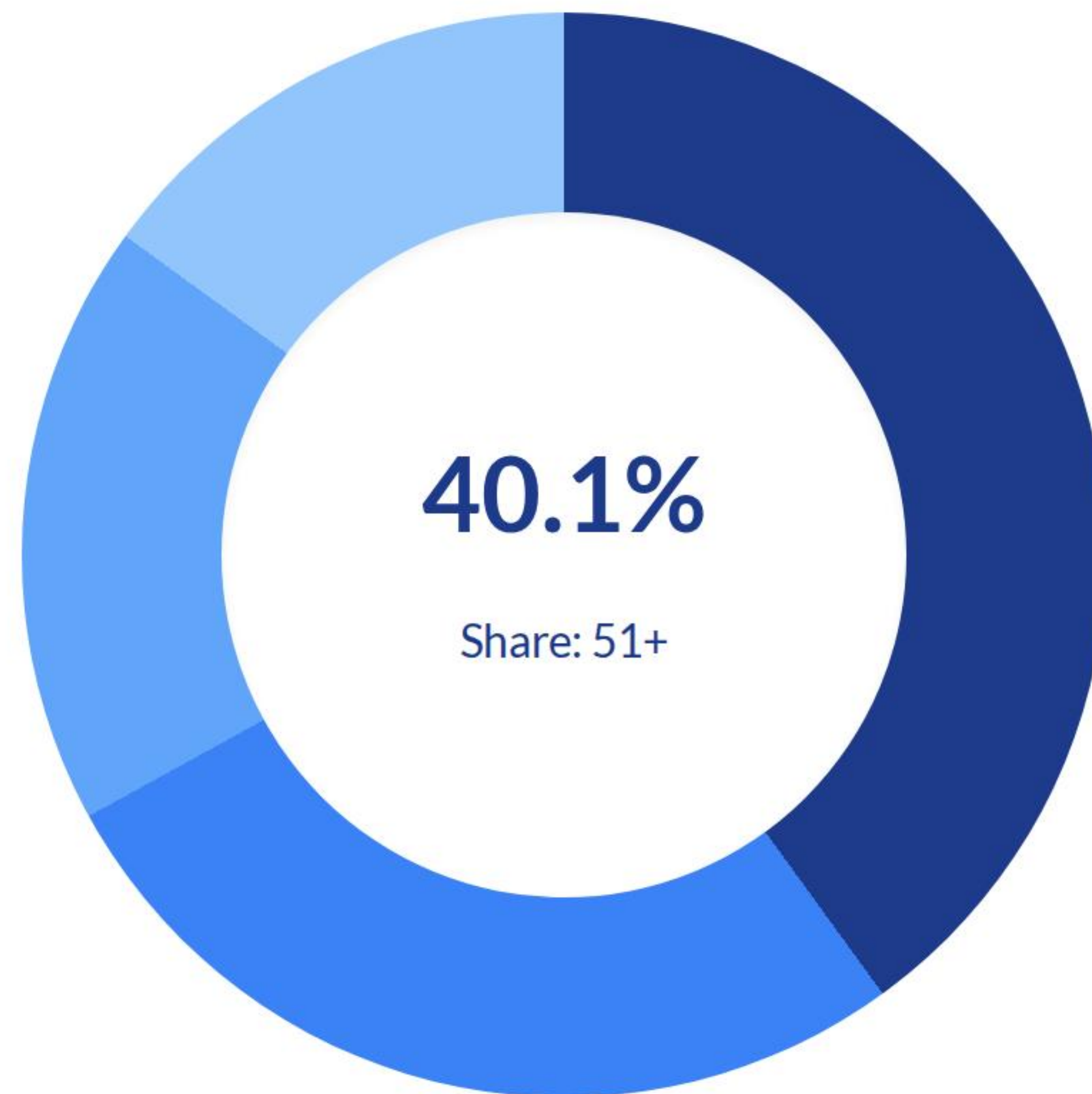


REVENUE BY DEPARTMENT



Electronics contributes nearly 50% of total portfolio revenue.

REVENUE BY AGE GROUP



- Age 51+ (\$387.6K)
- Age 36-50 (\$258.5K)
- Age 26-35 (\$173.0K)
- Age 18-25 (\$146.6K)

CUSTOMER SEGMENTATION

3,537

Loyal Customers

Purchased more than 10 times

Retention Pillars

Analysis of 5,000 customers identified through complex SQL CTEs:

Loyal: 70.7% (Power Users)

Returning: 18.1% (1-10 Purchases)

New: 11.2% (Growth Segment)

TOP CATEGORY DRIVERS

Department	Best Selling Item	Volume	Avg Rating
Accessories	Jewelry	171	3.76
Clothing	Shirt	315	3.43
Electronics	Phone	171	2.97
Footwear	Sandals	160	3.85
Outerwear	Jacket	166	3.76

PROMO STRATEGY ANALYSIS



Discount Revenue

\$474.8K

Significant portion of revenue is discount-driven.



Premium Cohort

441 Users

Used discounts but still spent above the mean average.

Strategic Alert: Electronics (Phones/Laptops) show >51% discount dependency, highlighting a need for value-based positioning.

LOGISTICS & SPENDING

The Speed Incentive

Logistics data confirms that premium shipping options attract high-value transactions.

Express: \$345.07 AOV

Standard: \$329.97 AOV

Subscription models should prioritize "Next Day" shipping to secure high-frequency buyers.



PATH TO GROWTH

- **Demographic Pivot:** Deploy targeted loyalty incentives for the 51+ Age Group.
- **Operational Tweak:** Bundle Express shipping with the Subscription tier to boost CLV.
- **Product Health:** Investigate quality issues in Electronics (low ratings) to drive non-promo sales.
- **Retention Play:** Use discount triggers to convert the 18% "Returning" segment into "Loyal."
- **Inventory Optimization:** Prioritize top-sellers (Jewelry, Shirts) in regional distribution centers.

Q&A

How can we transform this data into your next strategic win?

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IMAGE SOURCES



https://png.pngtree.com/thumb_back/fh260/background/20221015/pngtree-abstract-programming-workflow-on-a-screen-while-developing-real-python-code-photo-image_28458240.jpg

Source: [pngtree.com](https://png.pngtree.com)



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Source: www.freepik.com